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Ethnomedicinal studies on threatened, endangered plant species *Centella asiatica* Linn.

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Abstract : The paper concerns ethnomedicinal studies on endangered plant species of Jharkhand, *Centella asiatica* locally called "Beng sag" because of the shape of the leaves. It is one of the chief herbs for treating skin problems, for revitalizing the nerves and brain cells. *C.asiatica* is claimed to have preventive and curative properties against several diseases by the traditional herbal medicinal practitioners of Jharkhand.

Key Words: *Centella asiatica*, Ethnomedicine, Skin problems, Endangered, .

INTRODUCTION

Centella asiatica (Linn.) urban synonym *Hydrocotyle asiatica* (Linn.) commonly known as Indian Penny wort, belongs to the family Apiaceae (previously known as Umbelliferaece).[1,2,3] It is perennial herb with creeping and stoloniferous stem (Fig -A) . It is used to grow in the forest as well as in the kitchen gardens of Jharkhand and other parts of India also.[4,5,6]

The density of this species over the years have sharply declined because of its wide ethnomedicinal use. *C.asiatica* has been subjected to quite extensive experimental and clinical investigations particularly in the treatment of skin diseases and wound healing which has shown encouraging results. The whole plant is indiscriminately removed for use as ethnomedicinal and also for preparing a special juice from the paste of its leaves as brain food. For treating skin related problems, the tribals of Jharkhand apply the paste locally made of its tender leaves. The large scale and unrestricted exploitation of this natural resource results for its depletion and making it an endangered species.

MATERIAL AND METHOD .

From field survey it has been recorded that *Centella asiatica* is commonly known as 'beng sag' by the local tribals of Jharkhand. It is a creeper, faintly aromatic and valuable medicinal herb. *C.asitica* used to grow quite abundentaly few decades back as a wild species in the forest of Jharkhand, as the information gathered from the local tribal inhabitants and herbal medicine practitioner, however, at present it has become very rare species in the forest because of over exploitation . Some seedlings were carefully uprooted from the forest and transplanted in the garden of one of the senior herbal practitioner for futher studies and conservation with proper care.The transplanted plants showed good growth in the garden which served the material for present investigation.

OBSERVATION AND DISCUSSION

The tribals of Jharkhand use this plant against diverse diseases. *C.asitica* is used for the treatment of mental disorder, helminth diseases, blood impurity, jaundice and skin diseases etc. In addition, it is also used as a general tonic. The local tribal of Jharkhand use *C.asiatica* as a green vegetable in their general diet . Futhermore, they prepare powder of dried leaves of the plants and store the same for use in preparing curry. This may be one of the reasons why the diseases such as skin problems and jaundice is not as much prevalent amongst these people as other non-tribals in Jharkhand. Herbalist of Jharkhand often prescribe the use of *C.asiatica* to treat wounds, burns, cracks and minor skin lesions as the herb contains

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triterpenoids- substances believed to expedite wound healing by boosting antioxidants in the wound strengthening the skin and increasing blood supply to the wounded area . They also found that low doses of chemical compounds found in *C.asiatica* known as asiaticosides enhanced burn-wound healing.[7,8,9]

Requirement of *C.asitica* is rising sharply in view of the popularity of the mandukaparni based drugs.[10,11] Because of the large scale and unrestricted exploitation of this natural resources to meet ever increasing demand by the Indian Pharmaceutical industry which is coupled with the limited cultivation and insufficient attempts for its replenishments. *C.asiatica* was prevalent in the forest of Jharkhand few years back, however the population of this medicinal important plant species has been markedly depleted and it is listed as a threatened species by the Government as endangered species. Attempts for conservation of such species are most essential.

The wild stock of this medicinal important plant species has been markedly depleted and it is listed as a threatened species by International Union for conservation of Nature and Natural Resources[10] and an endangered species.[1,3]

In recent years , there has been an increasing interest in *in-vitro* culture techniques which offer a viable tool for mass multiplication and germplasm conservation of rare, endandered and threatened medicinal plants.[1,4,10] Therefore, it is important to develop an efficient micropropagation technique for *Centella asiatica* to rapidly disseminate superior clones once they are identified. Tissue culture technique can playan important role in the clonal and germplasm conservation.[1] In our country also there is a need to compile the information regarding traditional ethnomedicinal practices of different region so species may become endangered or extinct, and that the most suitable and efficacious drug may be obtain in future against different diseases.



Fig – A

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